

COMPUTER NETWORKS – HANDWRITTEN EXAM NOTES

UNIT 3: Medium Access Control (MAC), Random Access, ALOHA, CSMA/CD, CSMA/CA, Controlled Access, Channelization

B.Tech Mid-I Examination Preparation | Complete Question Bank Solutions

PART A – 2 MARKS QUESTIONS

Question 1

What is random access in the data link layer?

Definition & Key Characteristics

Random Access (Contention-based MAC) is a class of Multiple Access protocols where no single station is assigned a scheduled time slot or dedicated channel, and no station is superior to another.

- **Contention:** Any station that has data ready can attempt to transmit at any time.
- **Collisions:** If two or more stations transmit simultaneously on the shared medium, their signals overlap and collide, corrupting the data.
- **Key Protocols:** ALOHA (Pure & Slotted), CSMA, CSMA/CD, CSMA/CA.

Question 2

Define ALOHA

Definition

ALOHA is the earliest random access protocol, designed in 1970 by **Norman Abramson** at the University of Hawaii for ground-based packet radio networks.

- **Basic Premise:** A station transmits a frame whenever it has data ready. If an acknowledgment (ACK) is not received within a timeout period, the station assumes a collision occurred, waits a random backoff time, and retransmits.
- **Two Versions:** **Pure ALOHA** (continuous unslotted time, max throughput = 18.4%) and **Slotted ALOHA** (discrete synchronized time slots, max throughput = 36.8%).

Question 3

What is CSMA?

Definition & Principle

CSMA (Carrier Sense Multiple Access) improves upon ALOHA by following the fundamental principle: "Sense before Transmit" (or "Listen before Talk").

- Before transmitting, a node first senses the shared medium (carrier).
- If the channel is **busy**, the node defers transmission to avoid collision.
- If the channel is **idle**, the node proceeds to transmit based on its persistence strategy (1-persistent, non-persistent, or p-persistent).

Question 4

Differentiate between CSMA/CD and CSMA/CA

Feature	CSMA/CD (Collision Detection)	CSMA/CA (Collision Avoidance)
Core Strategy	Transmits and detects collisions while transmitting .	Takes proactive steps to avoid collisions before transmitting .
Reaction to Collision	Aborts immediately, sends a Jam signal , waits backoff time.	Waits for Interframe Space (IFS), uses Contention Window & Backoff.
Handshake	No handshake used.	Uses optional RTS/CTS (Request/Clear to Send) handshake.
Standard & Media	IEEE 802.3 Ethernet (Wired networks where listening while talking is possible).	IEEE 802.11 Wi-Fi (Wireless networks where full-duplex sensing is impractical).

Question 5

What is channelization? Name the channelization techniques

Definition

Channelization (Channel Partitioning) is a multiple access method in which the total available bandwidth of a shared link is divided simultaneously among multiple communicating stations along frequency, time, or code dimensions.

Three Primary Techniques

1. **FDMA (Frequency Division Multiple Access)**: Divides total bandwidth into distinct frequency bands; each station gets a dedicated frequency sub-band permanently.
2. **TDMA (Time Division Multiple Access)**: Entire bandwidth is allocated to one station at a time in periodic, repeating time slots.
3. **CDMA (Code Division Multiple Access)**: All stations transmit simultaneously over the entire frequency spectrum, separated by unique orthogonal mathematical codes (chip sequences).

----- **PART B – 5 MARKS / ESSAY QUESTIONS** -----

Essay Question 1

Explain the working of ALOHA and discuss its basic operation

1. Introduction to ALOHA

ALOHA is the pioneering random access protocol where stations share a common broadcast radio channel without central coordination.

2. Pure ALOHA

- **Operation:** Whenever a station generates a frame, it immediately transmits it.
- **Vulnerable Time (S_V):** A frame of transmission time T_{fr} collides if any other station transmits in the window from $t_0 - T_{fr}$ to $t_0 + T_{fr}$.

$$S_V = \text{Vulnerable Period (Pure ALOHA)} = 2 \times T_{fr}$$

- **Throughput Formula (S):** Assuming Poisson frame arrival rate G (offered load):

$$S = G \cdot e^{-2G}$$

- **Maximum Throughput:** Differentiating S with respect to G gives maximum at $G = 0.5$:

$$S_{\max} = \frac{1}{2e} \approx 0.184 = \mathbf{18.4\%}$$

3. Slotted ALOHA

- **Operation:** Time is divided into synchronized discrete slots of duration T_{fr} . A station can transmit only at the beginning of a time slot.
- **Vulnerable Time (S_V):** Collisions can occur only if two stations attempt to transmit in the exact same slot.

$$S_V = \text{Vulnerable Period (Slotted ALOHA)} = T_{fr}$$

- **Throughput Formula:**

$$S = G \cdot e^{-G}$$

- **Maximum Throughput:** Maximum occurs at $G = 1.0$:

$$S_{\max} = \frac{1}{e} \approx 0.368 = \mathbf{36.8\%} \text{ (Double of Pure ALOHA!)}$$

4. Summary Comparison: Pure vs Slotted ALOHA

Aspect	Pure ALOHA	Slotted ALOHA
Transmission Timing	Continuous (any time)	Synchronized at start of slot

Vulnerable Period	$2 T_{fr}$	T_{fr}
Throughput Formula	$S = G e^{-2G}$	$S = G e^{-G}$
Maximum Throughput	18.4% (at $G = 0.5$)	36.8% (at $G = 1.0$)
Clock Synchronization	Not required	Required (global clock sync)

Essay Question 2

Explain CSMA, CSMA/CD and CSMA/CA with suitable examples

1. CSMA (Carrier Sense Multiple Access) & Persistence Modes

In CSMA, a station senses the channel before transmitting. When the channel is busy, it employs one of three persistence strategies:

- 1-Persistent CSMA:** If channel is idle, transmit immediately with probability 1. If busy, keep sensing continuously until idle, then transmit. (High chance of collision if two stations wait for the same busy channel).
- Non-Persistent CSMA:** If channel is idle, transmit. If busy, wait for a **random amount of time** before sensing again. (Reduces collisions, but introduces idle channel delay).
- p-Persistent CSMA:** Used in slotted channels. If channel is idle, transmit with probability p ; with probability $(1-p)$, wait for next slot.

2. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

- Core Operation (Ethernet IEEE 802.3):** A station listens to the wire **while transmitting**. If the voltage level increases beyond a threshold, a collision is detected.
- Collision Action:** The station immediately aborts transmission, sends a 32-48 bit **Jam Signal** to notify all stations, and runs the **Binary Exponential Backoff Algorithm**:

Wait time $T = k \times \text{Slot Time}$, where k is chosen randomly from $[0, 2^i - 1]$ (for attempt $i \leq 10$).

- Minimum Frame Size Condition:** To ensure the sender detects collision before completing transmission, the frame transmission time T_{fr} must be at least twice the maximum propagation time T_p :

$$T_{fr} \geq 2 T_p \implies \frac{\text{Frame Length } (L)}{\text{Bandwidth } (B)} \geq 2 \times \frac{\text{Distance } (d)}{\text{Propagation Speed } (v)} \implies \mathbf{L_{min} = 2 \cdot B \cdot \frac{d}{v}}$$

3. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

- Why Collision Detection Fails in Wireless:** Wireless transceivers cannot transmit and receive on the same frequency simultaneously without self-interference drowning out distant collisions.
- Three Avoidance Mechanisms in IEEE 802.11:**

1. **Interframe Space (IFS):** When the channel becomes idle, the station waits for a period called **DIFS (Distributed IFS)** before doing anything.
2. **Contention Window & Backoff:** After DIFS, the station chooses a random backoff counter and decrements it only when the channel remains idle.
3. **RTS/CTS Handshake with NAV:** Sender transmits a small **RTS (Request to Send)** frame. Receiver responds with **CTS (Clear to Send)**. All other stations hearing RTS/CTS set a **Network Allocation Vector (NAV)** timer and remain silent, completely solving the **Hidden Station Problem!**

Essay Question 3

Explain controlled access techniques: Reservation, Polling and Token Passing

1. Concept of Controlled Access

In **Controlled Access**, stations consult one another to determine which station has the right to transmit. **No collisions can occur** because only one authorized station transmits at any given time.

2. The Three Controlled Access Protocols

(a) Reservation Method

- Time is organized into intervals. Each interval begins with a **Reservation Frame** consisting of N mini-slots (for N stations in the network).
- If station k wants to transmit data, it puts a 1 in the k -th mini-slot of the reservation frame.
- After the reservation frame passes, stations transmit their full data frames in the exact sequence in which reservations were made.

(b) Polling Method

- Works in a **Primary-Secondary (Master-Slave)** topology. One primary device controls the shared link, and secondary devices communicate only via the primary.
- **Poll Function:** Primary asks secondary: "Do you have data to send?" If yes, secondary sends data; if no, secondary sends NAK.
- **Select Function:** Primary tells secondary: "I have data for you. Are you ready to receive?"
- **Drawback:** High polling overhead and single point of failure at the primary.

(c) Token Passing Method

- Stations are arranged in a **Logical Ring** (Token Ring IEEE 802.5, FDDI).
- A special small control frame called the **Token** circulates endlessly around the ring from predecessor to successor.
- **Transmission:** If a station has data to send, it captures and holds the token, changes its state to a data frame, appends user data, and transmits it down the ring.
- **Token Release:** The destination reads the data, and when the frame circulates back to the sender, the sender strips the data and releases a new Token onto the ring for the next station.

- **Advantage:** Completely deterministic, bounded delay, fair channel access, zero collisions.

Essay Question 4

Explain channelization techniques: FDMA, TDMA and CDMA

1. Introduction to Channelization

Channelization is a multiple access mechanism where total transmission capacity is partitioned among stations in frequency, time, or mathematical code domains.

2. Detailed Channelization Protocols

(a) FDMA (Frequency Division Multiple Access)

- The total available bandwidth is divided into non-overlapping frequency bands.
- Each station is allocated a dedicated frequency sub-band for the entire duration of communication.
- **Guard Bands:** Small unused frequency strips placed between adjacent channels to prevent cross-channel interference.
- **Example:** Traditional analog AM/FM radio, 1G cellular networks (AMPS).

(b) TDMA (Time Division Multiple Access)

- All stations share the exact same frequency band, but transmit in dedicated, non-overlapping **time slots**.
- **Guard Times:** Small time buffers between time slots to account for propagation delay variations and clock synchronization drift.
- **Example:** 2G GSM cellular standard (8 time slots per carrier frequency).

(c) CDMA (Code Division Multiple Access)

- All stations transmit simultaneously across the entire frequency band.
- Each station is assigned a unique, orthogonal mathematical code vector called a **Chip Sequence** (Walsh Code).
- **Orthogonal Vector Properties:**

Let C_i and C_j be chip sequences of length N :

1. Inner product with itself: $C_i \cdot C_i = N$ ($\frac{1}{N} C_i \cdot C_i = 1$)
2. Inner product with another code: $C_i \cdot C_j = 0$ (Orthogonal!)

• Encoding & Decoding Example:

1. Data bit 0 is represented as -1 , data bit 1 is represented as $+1$, and silence is 0 .
2. Let Station 1 have code $C_1 = [+1, +1, +1, +1]$ and send data $d_1 = +1 \implies S_1 = [+1, +1, +1, +1]$.
3. Let Station 2 have code $C_2 = [+1, -1, +1, -1]$ and send data $d_2 = -1 \implies S_2 = [-1, +1, -1, +1]$.
4. Combined signal on the channel: $S = S_1 + S_2 = [0, +2, 0, +2]$.

5. Receiver for Station 1 extracts data:

$$\text{Data}_1 = \frac{1}{4} (S \cdot C_1) = \frac{1}{4} [(0 \times 1) + (2 \times 1) + (0 \times 1) + (2 \times 1)] = \frac{4}{4} = \mathbf{+1} \text{ (Bit 1 restored!)} \quad \square$$

3. Comprehensive Comparison Table

Parameter	FDMA	TDMA	CDMA
Division Basis	Frequency domain	Time domain	Mathematical Code domain
Frequency Sharing	Different frequencies	Same frequency	Same frequency simultaneously
Time Sharing	Simultaneous (all time)	Different time slots	Simultaneous (all time)
Synchronization	Simple (frequency filters)	Strict time sync required	Code chip synchronization
Capacity & Efficiency	Low	Moderate	Highest (Soft capacity limit)

----- LAST-MINUTE EXAM REVISION (UNIT 3) -----

Quick Master Cheat Sheet (Unit 3)

Protocol / Method	Key Formula / Characteristic	Exam Highlight
Pure ALOHA	$S = G e^{-2G}$, $S_{\max} = 18.4\%$	Vulnerable time $2 T_{fr}$
Slotted ALOHA	$S = G e^{-G}$, $S_{\max} = 36.8\%$	Vulnerable time T_{fr}
CSMA/CD Min Frame	$L_{\min} = 2 \cdot B \cdot (d/v)$	$T_{fr} \geq 2 T_p$
CSMA/CA	DIFS, Backoff, RTS/CTS, NAV	Avoids Hidden Station Collisions
Token Passing	Deterministic ring token	Zero collisions, fair access
CDMA Walsh Codes	$C_i \cdot C_j = 0$, $C_i \cdot C_i = N$	All stations transmit on same frequency & time